

TRB Annual Meeting

Understanding Accessibility Sufficiency, Activity Participation, and Quality of Life: Evidence from Austin, Texas --Manuscript Draft--

Full Title:	Understanding Accessibility Sufficiency, Activity Participation, and Quality of Life: Evidence from Austin, Texas
Abstract:	Sufficient accessibility means that all individuals have a minimum level of access to essential opportunities such as work, school, healthcare, and leisure, making it possible to meet their basic needs. Using a combination of accessibility estimates, travel-activity survey data, and a local community survey, we aim to identify a sufficient accessibility level that ensures basic activity needs and facilitates a higher quality of life (QoL). We used both primal and dual accessibility estimates to create a composite accessibility index that can identify multimodal accessibility levels in Austin, TX. Our findings reveal a significant disparity in auto and transit access to jobs and other non-work destinations like parks, grocery stores, and hospitals. Our inequality measures, including the Gini coefficient, Lorenz curve, and Concentration Index (CI), highlight inequality in accessibility distributions, favoring lower-income groups. Despite higher accessibility in central regions, lower-income households without vehicles participate in fewer activities. We used a multivariate model to estimate the impact of the composite accessibility index on household activity participation. We found a moderate but significant positive effect of composite accessibility on daily activity participation rates. Also, significant differences in average activity participation and accessibility scores were observed across different QoL categories, suggesting an insufficient accessibility level for the disadvantaged population. We measured the depth and severity of insufficient accessibility using the Foster-Greer-Thorbecke (FGT) index. Our study underscores the need for targeted interventions to support lower-income, carless households and calls for policies addressing economic barriers to translate higher accessibility into improved activity participation and QoL.
Additional Information:	
Question	Response
The total word count limit is 7500 words including tables. Each table equals 250 words and must be included in your count. Papers exceeding the word limit may be rejected. My word count is:	6989
Manuscript Classifications:	Data and Data Science; Geographic Information Science AED40; Geographic Information Systems; Planning and Analysis; Transportation Planning Analysis and Application AEP15; Accessibility Planning; Geographic Information Systems (GIS) Planning Analysis and Application; Transportation Planning Policy and Processes AEP10; Access; Sustainability and Resilience; Transportation and Society; Community Resources and Impacts AME80; Accessibility; Equity in Transportation AME10; Accessibility
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ABSTRACT

Sufficient accessibility means that all individuals have a minimum level of access to essential opportunities such as work, school, healthcare, and leisure, making it possible to meet their basic needs. Using a combination of accessibility estimates, travel-activity survey data, and a local community survey, we aim to identify a sufficient accessibility level that ensures basic activity needs and facilitates a higher quality of life (QoL). We used both primal and dual accessibility estimates to create a composite accessibility index that can identify multimodal accessibility levels in Austin, TX. Our findings reveal a significant disparity in auto and transit access to jobs and other non-work destinations like parks, grocery stores, and hospitals. Our inequality measures, including the Gini coefficient, Lorenz curve, and Concentration Index (CI), highlight inequality in accessibility distributions, favoring lower-income groups. Despite higher accessibility in central regions, lower-income households without vehicles participate in fewer activities. We used a multivariate model to estimate the impact of the composite accessibility index on household activity participation. We found a moderate but significant positive effect of composite accessibility on daily activity participation rates. Also, significant differences in average activity participation and accessibility scores were observed across different QoL categories, suggesting an insufficient accessibility level for the disadvantaged population. We measured the depth and severity of insufficient accessibility using the Foster-Greer-Thorbecke (FGT) index. Our study underscores the need for targeted interventions to support lower-income, carless households and calls for policies addressing economic barriers to translate higher accessibility into improved activity participation and QoL.

Keywords: Accessibility, Activity Participation, Quality of Life

1 is also applicable for the community survey from which we derived QoL. Community survey
2 data included few transports sector specific satisfaction related questions i.e., traffic flow in
3 major roads, enforcement of traffic laws, conditions of sidewalks and city streets, pedestrian
4 accessibility, mode choice options, timing of traffic signals etc. Due to data availability issue in
5 recent version of the survey, we wouldn't include those indicators in overall well-being measure.
6 Future work should include a comprehensive formulation of subjective well-being measures
7 along with objectively measured aspects of well-being, such as accident fatalities, crime rates,
8 housing affordability, air quality, and physical activity levels. This approach would help to
9 determine a sufficient accessibility threshold.

10 **ACKNOWLEDGMENTS**

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13 of Transportation.

15 **AUTHOR CONTRIBUTIONS**

16 Authors confirm contribution to the paper as follows: study conception and design: all; data
17 collection: Rahman; analysis and interpretation of results: all; draft manuscript preparation: all.
18 All authors reviewed the results and approved the final version of the manuscript.